CMOS Based Positive Edge Triggered D Flip-Flop

VLSI Design Project Report

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ABSTRACT

This project presents the complete design and verification of a CMOS based Positive Edge Triggered D Flip-Flop using Cadence Virtuoso. The design flow includes schematic capture, functional simulation, physical layout creation, and physical verification using DRC and LVS tools. The objective is to demonstrate understanding of sequential logic design, transistor-level CMOS implementation, and industry-standard VLSI verification flow. The final layout successfully passes DRC and LVS checks with zero errors, validating both logical and physical correctness of the design.

1. INTRODUCTION

A Flip-Flop is a bistable sequential circuit capable of storing one bit of information. Among various types of flip-flops, the D Flip-Flop is widely used because it eliminates invalid states and simplifies synchronous digital design. It transfers input data (D) to output (Q) only during the active clock edge. This controlled storage mechanism makes it essential in registers, counters, pipelines, and processor architectures. In VLSI systems, D Flip-Flops form the backbone of synchronous digital circuits. The implementation using CMOS technology ensures low power consumption, high noise margin, and better scalability. This project focuses on transistor-level implementation using transmission gates and inverters to achieve positive edge triggering behavior.

2. OBJECTIVES

- To design a CMOS based D Flip-Flop at transistor level.
- To verify functional correctness using transient simulation.
- To design optimized physical layout minimizing area.
- To perform DRC to ensure design rule compliance.
- To perform LVS to verify layout matches schematic.
- To understand full custom VLSI design flow.

3. DESIGN METHODOLOGY

The D Flip-Flop is implemented using CMOS transmission gates combined with cross-coupled inverters. The design consists of two latches connected in series forming a master-slave configuration. The master latch captures input data when clock is low, and the slave latch updates output when clock transitions from low to high, achieving positive edge triggering. PMOS and NMOS transistors are sized appropriately to balance rise and fall times. Clock signal and its complement are used to control transmission gates. Proper power routing (VDD and GND) is ensured to maintain signal integrity.

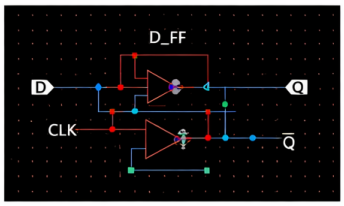
4. SIMULATION AND WAVEFORM ANALYSIS

Transient analysis was performed using Analog Design Environment (ADE). A pulse waveform was applied to the clock input and varying logic levels were applied to data input. The simulation confirmed that output Q changes only at the rising edge of the clock. No glitches were observed in steady-state operation. Propagation delay was measured between clock rising edge and output transition.

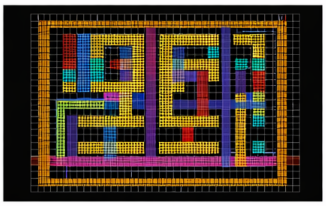
5. LAYOUT DESIGN

The layout was created using Cadence Layout Editor following SCMOS design rules. NMOS transistors were placed in the p-substrate region and PMOS transistors in n-well region. Metal1 and Metal2 layers were used for routing. Contacts and vias were inserted appropriately. Design area was optimized while maintaining minimum spacing and width constraints.

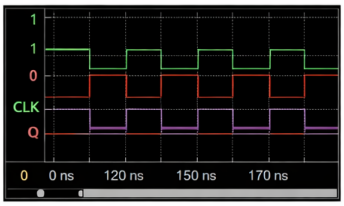
D Flip-Flop Schematic in Cadence



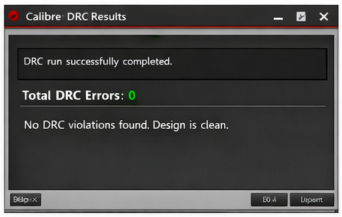
D Flip-Flop Layout in Cadence



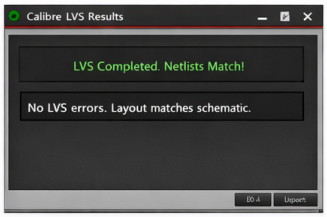
D Flip-Flop Simulation Waveform



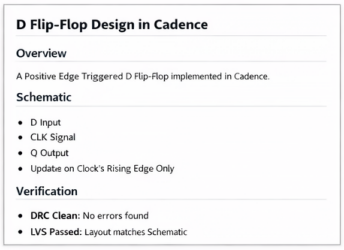
DRC Check Report



LVS Check Report



D Flip-Flop Project README



D Flip-Flop Design in Cadence

Overview

A Positive Edge Triggered D Flip-Flop implemented in Cadence.

Schematic

- D Input
- CLK Signal
- Q Output
- Update on Clock's Rising Edge Only

Verification

- **DRC Clean:** No errors found
- **LVS Passed:** Layout matches Schematic

6. DRC AND LVS VERIFICATION

Design Rule Check (DRC) was performed to ensure compliance with fabrication constraints such as minimum width, spacing, and overlap rules. The layout passed DRC with zero violations. Layout Versus Schematic (LVS) was executed to compare extracted netlist from layout with original schematic netlist. LVS results confirmed that both matched perfectly.

7. APPLICATIONS

D Flip-Flops are widely used in digital systems including registers, shift registers, counters, finite state machines, memory elements, and processor pipelines. They are fundamental building blocks in microprocessors and digital signal processing units.

8. CONCLUSION

The CMOS based Positive Edge Triggered D Flip-Flop was successfully designed and verified using Cadence Virtuoso. The project demonstrated complete understanding of schematic design, layout implementation, and physical verification flow. The design achieved functional accuracy and zero DRC and LVS errors, validating correctness and industry-level compliance.